

PINK REPRODUCTION

Reproducing PINK, Figure by Figure

Liu et al., J. Mater. Inf. 2025, 5, 12

every figure of the paper, our version beside it – and the ones we cannot
produce

PART ONE

Figure 1 --- the pipeline, and its physics

Eight equations, every symbol with its unit

01 · FIGURE 1

The workflow, end to end



Blue – learned. The only fitted part of the whole chain. A graph network maps structure to two elastic moduli.

Amber – physics. Closed form, no free parameters, no fitting. Equations 1-8 on the next three slides.

The network never sees a thermal conductivity. Every structural quantity (V , ρ , N , $\bar{M}$) is read from the CIF, not predicted – which is why two of our screens can differ only through K and G .

Equations 1--3 --- from moduli to sound

The two predicted moduli fix how fast sound travels through the crystal, and phonon speed is what sets thermal conductivity.

$$v_l = \sqrt{\frac{K + \frac{4}{3}G}{\rho}} \quad \text{longitudinal wave speed} \quad (1)$$

$$v_t = \sqrt{\frac{G}{\rho}} \quad \text{transverse wave speed} \quad (2)$$

$$v_s = \left[\frac{1}{3} \left(\frac{1}{v_l^3} + \frac{2}{v_t^3} \right) \right]^{-1/3} \quad \text{averaged sound velocity} \quad (3)$$

symbol	meaning	unit
K	bulk modulus (<i>predicted</i>)	GPa
G	shear modulus (<i>predicted</i>)	GPa
ρ	mass density (<i>from the CIF</i>)	g cm^{-3}

Equations 4--6 --- Debye temperature and Grüneisen

$$\Theta_D = \frac{h}{k_B} \left(\frac{3}{4\pi V} \right)^{1/3} v_s \quad \text{Debye temperature} \quad (4)$$

$$\nu = \frac{(v_l/v_t)^2 - 2}{2(v_l/v_t)^2 - 2} \quad \text{Poisson's ratio} \quad (5)$$

$$\gamma = \frac{3(1 + \nu)}{2(2 - 3\nu)} \quad \text{Grüneisen parameter} \quad (6)$$

sym	meaning	unit	sym	meaning	unit
h	Planck const.	J s	Θ_D	Debye temp.	K
k_B	Boltzmann const.	J K ⁻¹	ν	Poisson's ratio	–
V	vol. per atom (CIF)	Å ³	γ	Grüneisen	–

Eq. 6 matters most: γ is derived from the same K , G .

Equations 7--8 --- two routes to κ_L

$$\kappa_L^{\text{Slack}} = \frac{A(\gamma) \bar{M} V^{1/3} \Theta_D^3}{\gamma^2 T N}, \quad A(\gamma) = \frac{2.43 \times 10^{-8}}{1 - \frac{0.514}{\gamma} + \frac{0.228}{\gamma^2}} \quad (7)$$

$$\kappa_L^{\text{cal}} = \frac{G v_s V^{1/3}}{N T} e^{-\gamma} \quad (\text{PINK Eq. 2 - three-phonon, } \delta = 1) \quad (8)$$

symbol	meaning	unit
$\bar{M}$	mean atomic mass	amu
N	atoms per unit cell	–
T	temperature, fixed at 300	K
$A(\gamma)$	Slack's empirical prefactor	–
κ_L	lattice thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$

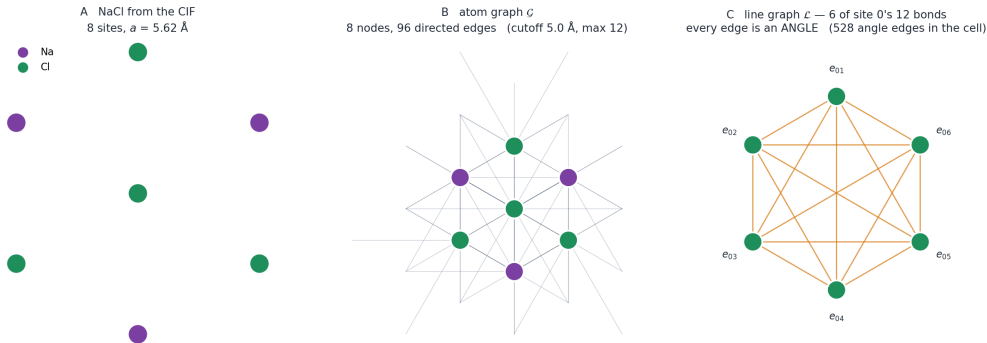
Both are implemented in `07_predict_kappa.py` and checked against the paper's own released code. **This project reports κ_L^{cal}** , so note its γ dependence is *exponential* – not the γ^{-2} of the Slack form. Substituting a different γ rescales it by $e^{\gamma_{\text{old}} - \gamma_{\text{new}}}$.

PART TWO

From CIF to two graphs

Computed from the real NaCl file, not sketched

NaCl, all the way through



Generated by `make_pink_figures.py` from `nacl.cif` at **ALIGNN's own graph settings** – cutoff $5.0 \text{ \AA}$, max 12 neighbours, read from the trained model's config rather than chosen for the picture. Projection is isometric: in rock salt every (x, y) holds both a Na and a Cl differing only in z , so a flat projection hides one species entirely.

What the three panels cost

The counts on the previous slide are computed, and they show why the line graph is the expensive half of ALIGNN:

NaCl, one unit cell

atoms (nodes of $\mathcal{G}$) 8

bonds (edges of $\mathcal{G}$) 96

angles (edges of $\mathcal{L}$) 528

Each atom keeps its 12 nearest neighbours, so it contributes 12 bond-nodes and $\binom{12}{2} = 66$ angle edges. Over 8 atoms that is $8 \times 66 = 528$ – **5.5× more edges than the atom graph has.**

Why promote bonds to nodes at all

A message-passing layer updates a node from its neighbours. On the atom graph the only geometric quantity a message can carry is the bond *length* – an angle belongs to a *pair* of bonds, and there is nowhere on $\mathcal{G}$ for it to live.

In $\mathcal{L}$ a pair of bonds *is* an edge, and an edge is exactly where a feature attaches.

Angles enter as $\cos \theta_{jik}$ on a radial basis, the same way distances do – smooth where it matters, no discontinuity at 0 or π .

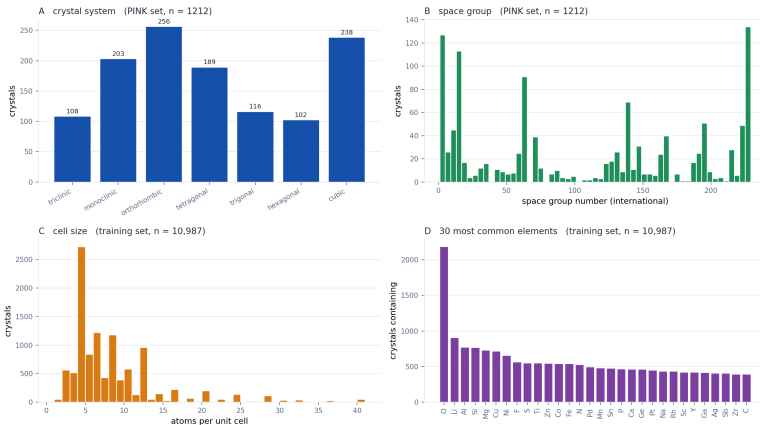
Shear modulus is a bond-bending property. CGCNN infers it from lengths alone; ALIGNN is handed the angles directly.

PART THREE

Figures 4, 5 and Table 1

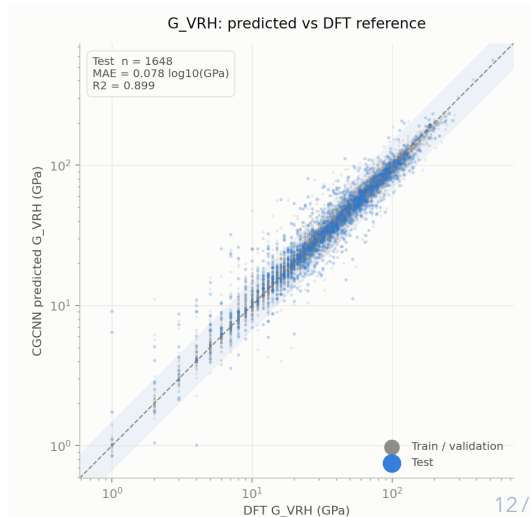
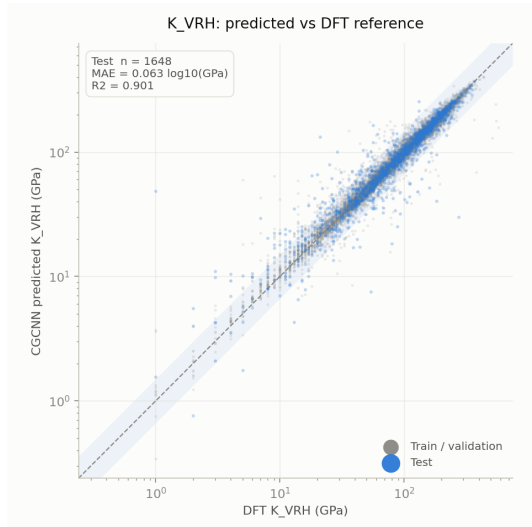
The data, the fit, and the measured check

Dataset statistics

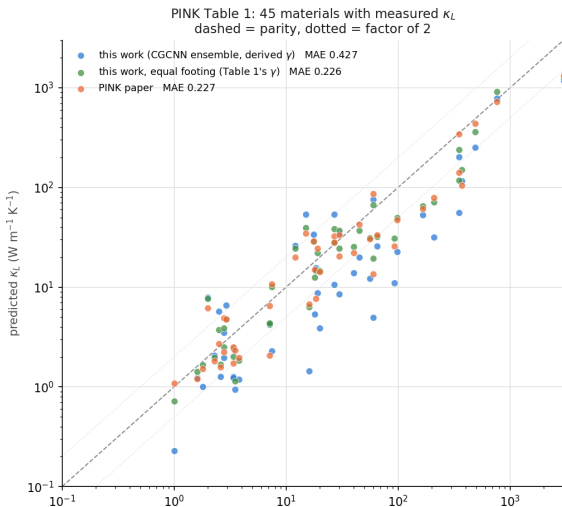


Each panel is titled with the population it describes. A/B use the 1,213 CIFs we hold; the 10,987 training structures are not available offline, so C/D use formulas and atom counts. Cubic dominates, as in the paper.

Predicted vs DFT --- the moduli



45 materials with measured κ_L



The paper's Table 1 lists 46 materials with **experimentally measured** κ_L ; 45 survive (mp-3490/GaP is deprecated in Materials Project).

mean $|\Delta \log_{10}|$

ours, derived γ	0.427
ours, equal footing	0.226
the PINK paper	0.227

Table 1's γ comes from AFLOW/experimental lookups, not from Eq. 6. Using Eq. 6 on both sides – equal footing – we match the paper.

`results/cgcnn/63_experimental_vs_predicted.csv`

45 rows, 27 columns.

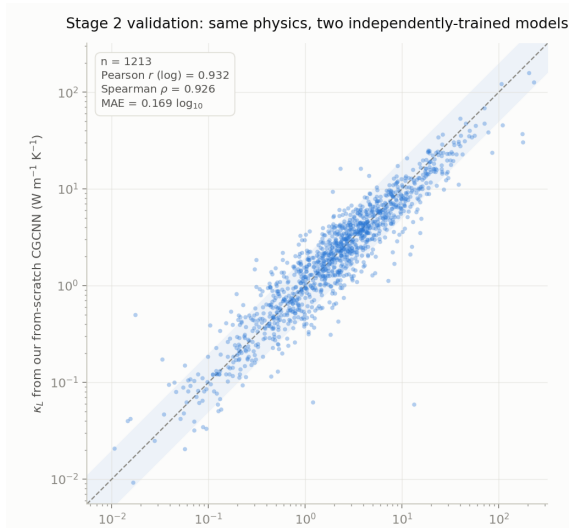
PART FOUR

Figures 6, 7 and 8

Validation, the funnel, and what came out of it

10 • FIGURE 6

κ_L against an independent pipeline



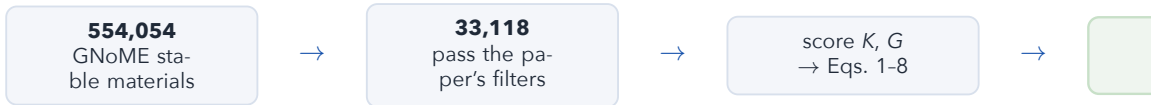
Our κ_L against the paper's own pretrained pipeline on the same 1,213 crystals:
Pearson $r = 0.932$, Spearman $\rho = 0.926$.

The equivalent architecture check –
ALIGNN's moduli through the *unchanged* physics – agrees even more tightly with our CGCNN ensemble: $r = 0.940$, $\rho = 0.947$.

Two independently trained architectures, sharing only the data, the split and Equations 1-8, converging this closely is evidence the pipeline finds physical signal rather than architecture-specific noise.

11 • FIGURE 7

The screening funnel



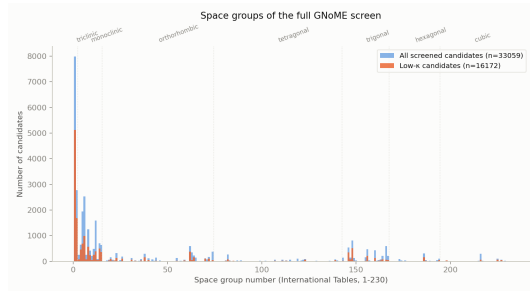
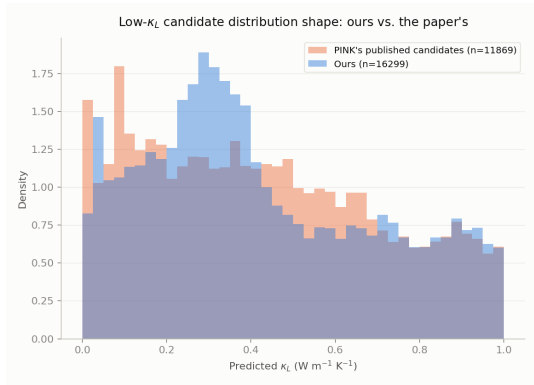
The paper's funnel reads 377,221 → 30,199 → 26,305 → 11,869. Ours starts from a larger number because the GNoME snapshot grew after publication – stated rather than glossed, because it means the two candidate lists are not directly comparable.

70.8% of the paper's own 11,869 published candidates also appear in our independently derived list.

Flagged: that figure predates several later corrections and should be re-computed before publication.

12 • FIGURE 8

What the candidates look like



B space groups of the screened pool

A κ_L distribution, ours vs the paper's

Both distributions span the same range and peak in the same place. Our candidate set is somewhat more concentrated around $0.25\text{--}0.4 \text{ W m}^{-1} \text{K}^{-1}$.

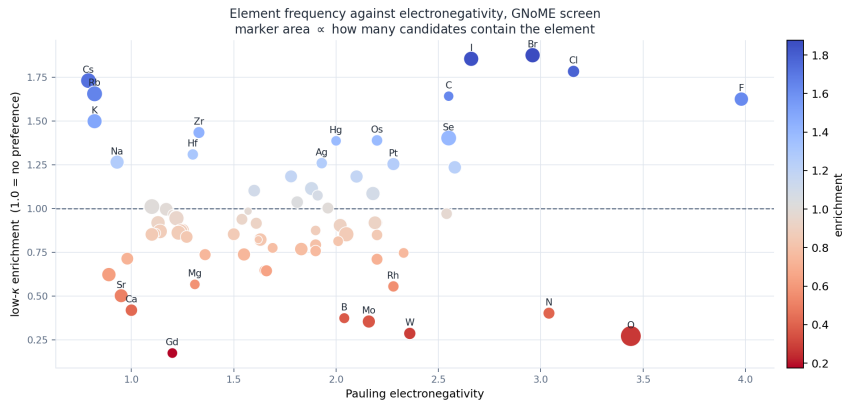
PART FIVE

Classification of the screen

By element, symmetry, family, and what we chose for DFT

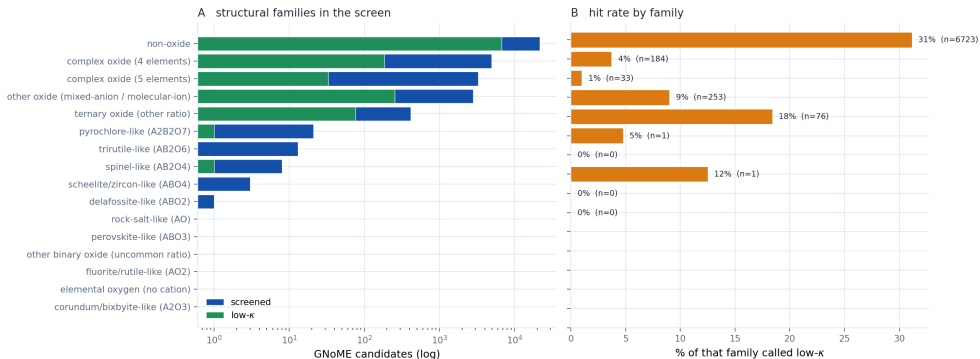
13 · CLASSIFICATION

Elements, against electronegativity



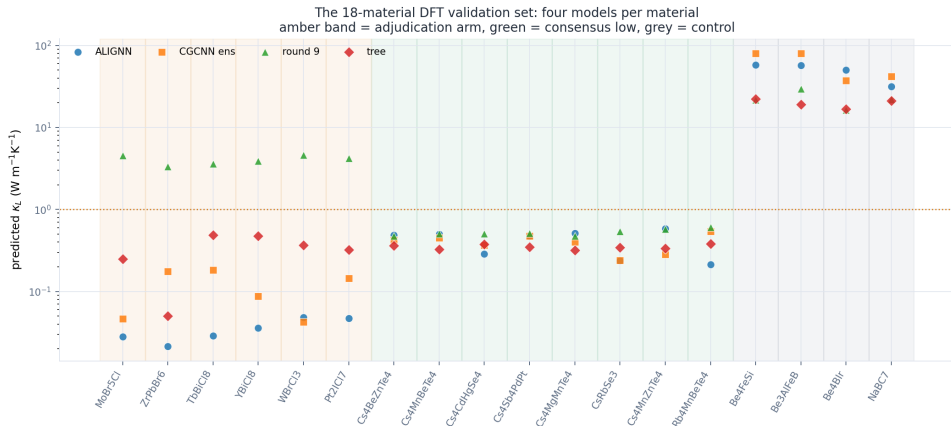
Enrichment is the rate at which candidates containing an element are called low- κ , divided by the base rate. Above 1.0 the element prefers the low- κ set. **Halogens sit high and to the right; oxygen sits lowest** – heavy soft halide lattices against tight oxide bonding, which is what Equations 1–3 predict, since a low G means a low v_s means a low κ_L .

Structural families



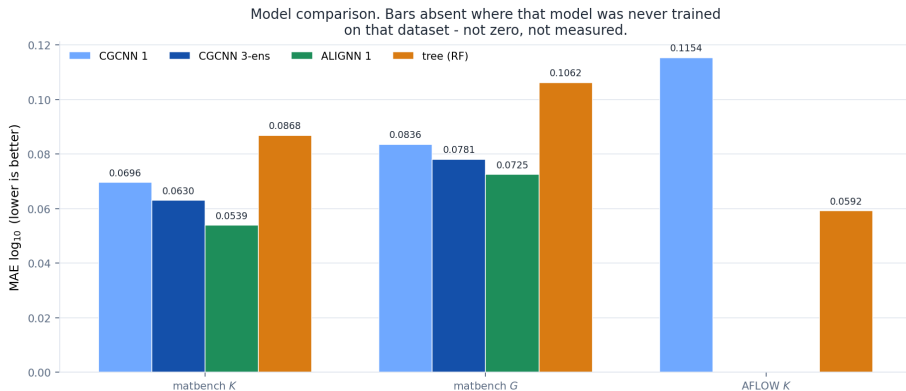
From `results/families/40_family_summary.csv`. Non-oxides dominate the screen and have by far the highest hit rate; complex 5-element oxides are numerous but almost never low- κ – **33 of 3,296**. GNoME’s discovered materials skew toward complex, previously uncatalogued compositions, so simple stoichiometries are largely absent.

The DFT validation set



18 materials, three arms, all ≤ 10 atoms. **A** consensus low (all four models agree) **B** adjudication (three say low, round 9 says high – these settle a disagreement nothing else can) **C** negative control (all four say stiff). Start with **MoBr₅Cl**: 7 atoms, *Cm*, ALIGNN 0.028 against round 9's 4.524.

Every model this project has trained



Bars are absent where a model was never trained on that dataset – not zero, not measured. **The composition-only tree beats the CGCNN on AFLOW** (0.0592 vs 0.1154) while losing on matbench; that is underfitting, not a limit of graph networks – the network’s *training* error on AFLOW (0.103) exceeds the tree’s *test* error.

FIGURES 9 AND 10

What we cannot reproduce

The paper's last two figures are **DFT phonon calculations** on $\text{Ag}_3\text{Te}_4\text{X}$ – the primitive structures, phonon dispersions, and then κ_L against temperature, cumulative κ_L , specific heat, mode group velocities, scattering phase space and scattering rates.

Producing them needs a full DFT plus finite-displacement phonon workflow with **third-order force constants**, on materials this project has never computed. That is days of DFT per material and an entirely different toolchain from anything here.

No stand-in is drawn for them. They are the same missing ground-truth gap the 18-material DFT validation set exists to start closing – and the elastic tensors in that set are the cheap first step toward exactly this kind of calculation.